

WEEK 7 PRACTICAL

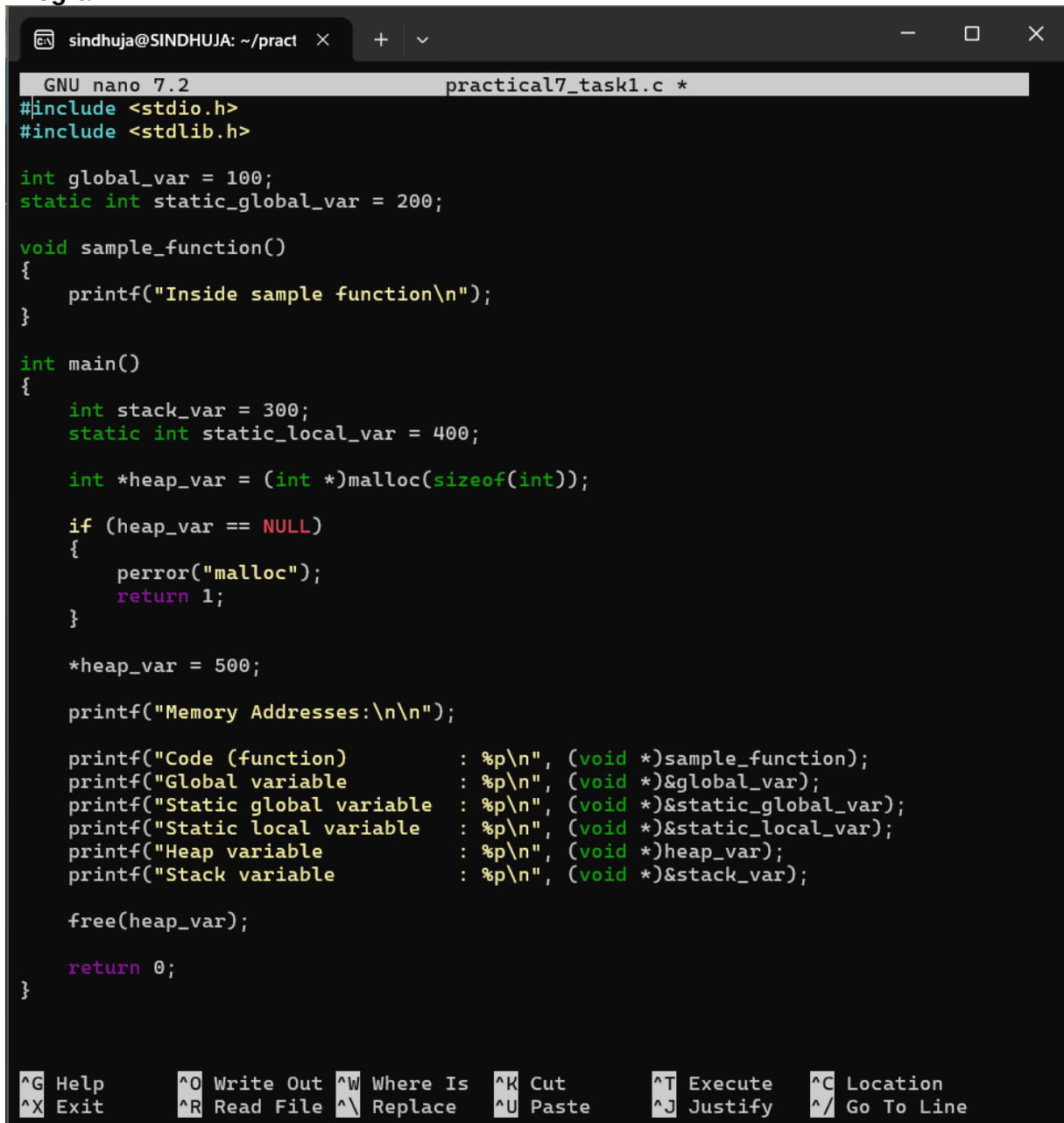
QUESTION 1

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Aim

To write a C program that displays the memory addresses of variables belonging to the **code**, **global**, **static**, **heap**, and **stack** segments and understand the Linux process address-space layout.

Program



```
GNU nano 7.2 practical7_task1.c *
#include <stdio.h>
#include <stdlib.h>

int global_var = 100;
static int static_global_var = 200;

void sample_function()
{
    printf("Inside sample function\n");
}

int main()
{
    int stack_var = 300;
    static int static_local_var = 400;

    int *heap_var = (int *)malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 500;

    printf("Memory Addresses:\n\n");

    printf("Code (function)      : %p\n", (void *)sample_function);
    printf("Global variable         : %p\n", (void *)&global_var);
    printf("Static global variable      : %p\n", (void *)&static_global_var);
    printf("Static local variable       : %p\n", (void *)&static_local_var);
    printf("Heap variable               : %p\n", (void *)heap_var);
    printf("Stack variable              : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Output

The exact addresses will be different on every execution because of Linux memory-layout randomization.

Memory Addresses:

```
sindhuja@SINDHUJA: ~/pract x + v
sindhuja@SINDHUJA:~$ cd ~
sindhuja@SINDHUJA:~$ rm -rf practical_week7
sindhuja@SINDHUJA:~$ ls
OSSP          practical4_task2_zombie  process        task3
client        practical4_task2_zombie.c process.c       task4
client.c      practical_week3         skill_week4    task5
fork_exec     practical_week4         skill_week5    task5.c
fork_exec.c   practical_week5         task1
linux_commands practical_week6         task1.c
sindhuja@SINDHUJA:~$ mkdir practical_week7
sindhuja@SINDHUJA:~$ cd practical_week7
sindhuja@SINDHUJA:~/practical_week7$ nano practical7_task1.c
sindhuja@SINDHUJA:~/practical_week7$ gcc practical7_task1.c -o practical7_task1
sindhuja@SINDHUJA:~/practical_week7$ ./practical7_task1
Memory Addresses:

Code (function)      : 0x60113b4f31e9
Global variable      : 0x60113b4f6010
Static global variable : 0x60113b4f6014
Static local variable : 0x60113b4f6018
Heap variable        : 0x60116c4512a0
Stack variable       : 0x7ffecb5e013c
sindhuja@SINDHUJA:~/practical_week7$ |
```

Observation

The program displays the addresses of variables located in different parts of the process address space.

The addresses demonstrate that different types of data are maintained in different regions of the Linux process address space.

Result

The program was successfully implemented to print the addresses of **code, global, static, heap, and stack variables**. The Linux process memory layout was analyzed based on the addresses obtained.

QUESTION 2:

Make use of the /proc/maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Aim

To study the memory organization of a running Linux process using the /proc/<PID>/maps file and memory-analysis commands, and understand virtual memory mappings.

Program

```
sindhuja@SINDHUJA: ~/pract × + v
GNU nano 7.2 practical7_task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

int main()
{
    int local_var = 200;

    printf("Process ID: %d\n", getpid());
    printf("Global variable address: %p\n", (void *)&global_var);
    printf("Stack variable address : %p\n", (void *)&local_var);

    printf("\nProcess is running...\n");
    printf("Open another terminal and execute:\n");
    printf("cat /proc/%d/maps\n", getpid());

    printf("\nPress Enter to terminate...\n");
    getchar();

    return 0;
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

Output

```
sindhuja@SINDHUJA:~/practical_week7$ nano practical7_task2.c
sindhuja@SINDHUJA:~/practical_week7$ gcc practical7_task2.c -o practical7_task2
sindhuja@SINDHUJA:~/practical_week7$ ./practical7_task2
Process ID: 535
Global variable address: 0x5a04c3925010
Stack variable address : 0x7ffe79313b74

Process is running...
Open another terminal and execute:
cat /proc/535/maps

Press Enter to terminate...
|
```

Useful Memory Analysis Commands

To examine the process mappings:

cat /proc/1722/maps

Sample /proc/1722/maps Output

```
sindhuja@SINDHUJA:~/OSSP/Week7$ cat /proc/1722/map
5ee50db78000-5ee50db79000 r--p 00000000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
5ee50db79000-5ee50db7a000 r-xp 00001000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
5ee50db7a000-5ee50db7b000 r--p 00002000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
5ee50db7b000-5ee50db7c000 r--p 00002000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
5ee50db7c000-5ee50db7d000 rw-p 00003000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
5ee533342000-5ee533363000 rw-p 00000000 00:00 0 [heap]
74a7ba608000-74a7ba628000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
74a7ba628000-74a7ba7c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
74a7ba7c0000-74a7ba80e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
74a7ba80e000-74a7ba812000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
74a7ba812000-74a7ba814000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
74a7ba814000-74a7ba821000 rw-p 00000000 00:00 0
74a7ba8df000-74a7ba8e2000 rw-p 00000000 00:00 0
74a7ba8e8000-74a7ba8ea000 rw-p 00000000 00:00 0 [vvar]
74a7ba8ea000-74a7ba8ee000 rw-p 00000000 00:00 0 [vvar_vclock]
74a7ba8ee000-74a7ba8f0000 rw-p 00000000 00:00 0 [vdso]
74a7ba8f0000-74a7ba8f2000 r-xp 00000000 00:00 0
74a7ba8f2000-74a7ba8f3000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74a7ba8f3000-74a7ba922000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74a7ba922000-74a7ba92d000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74a7ba92d000-74a7ba92f000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74a7ba92f000-74a7ba930000 r--p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74a7ba930000-74a7ba931000 rw-p 00000000 00:00 0
7ffd2f638000-7ffd2f651000 rw-p 00000000 00:00 0 [stack]
sindhuja@SINDHUJA:~/OSSP/Week7$
```

To view memory statistics:

cat /proc/1722/status

```
sindhuja@SINDHUJA:~/OSSP/Week7$ cat /proc/1722/status
Name: memory_maps
Umask: 0022
State: S (sleeping)
Tgid: 1722
Ngid: 0
Pid: 1722
PPid: 793
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1000 1000 1000 1000
FDSize: 256
Groups: 4 24 27 30 46 100 1000
NStgid: 1722
NSpid: 1722
NSpgid: 1722
NSsid: 793
Kthreadd: 0
VmPeak: 2768 kB
VmSize: 2768 kB
VmLck: 0 kB
```

To view detailed memory information:

cat /proc/1722/smaps

```
sindhuja@SINDHUJA:~/OSSP/Week7$ cat /proc/1722/smaps
5ee50db78000-5ee50db79000 r--p 00000000 08:30 80251 /home/sindhuja/OSSP/Week7/memory_maps
Size: 4 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 0 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 4 kB
Private_Dirty: 0 kB
Referenced: 4 kB
Anonymous: 0 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
Shared_Hugetlb: 0 kB
Private_Hugetlb: 0 kB
Swap: 0 kB
SwapPss: 0 kB
Locked: 0 kB
THPeligible: 0
```

You can also use:

pmap 1722

```
sindhuja@SINDHUJA:~/OSSP/Week7$ pmap 1722
1722: ./memory_maps
00005ee50db78000 4K r---- memory_maps
00005ee50db79000 4K r-x-- memory_maps
00005ee50db7a000 4K r---- memory_maps
00005ee50db7b000 4K r---- memory_maps
00005ee50db7c000 4K rw--- memory_maps
00005ee535342000 132K rw--- [ anon ]
000074a7ba608000 160K r---- libc.so.6
000074a7ba628000 1632K r-x-- libc.so.6
000074a7ba7c0000 312K r---- libc.so.6
000074a7ba80e000 16K r---- libc.so.6
000074a7ba812000 8K rw--- libc.so.6
000074a7ba814000 52K rw--- [ anon ]
000074a7ba8df000 12K rw--- [ anon ]
000074a7ba8e8000 8K rw--- [ anon ]
000074a7ba8ea000 16K r---- [ anon ]
000074a7ba8ee000 8K r---- [ anon ]
000074a7ba8f0000 8K r-x-- [ anon ]
000074a7ba8f2000 4K r---- ld-linux-x86-64.so.2
000074a7ba8f3000 188K r-x-- ld-linux-x86-64.so.2
000074a7ba922000 44K r---- ld-linux-x86-64.so.2
000074a7ba92d000 8K r---- ld-linux-x86-64.so.2
000074a7ba92f000 4K rw--- ld-linux-x86-64.so.2
000074a7ba930000 4K rw--- [ anon ]
00007ffd2f630000 132K rw--- [ stack ]
total 2768K
sindhuja@SINDHUJA:~/OSSP/Week7$
```

Observation

The `/proc/<PID>/maps` file shows the **virtual memory regions** assigned to a running process.

Common mappings include:

- **Executable/text region** — contains the program's machine instructions.
- **Read-only region** — contains read-only data and other protected mappings.
- **Data region** — contains initialized and writable program data.

- **Heap** — used for dynamically allocated memory.
- **Shared libraries** — libraries such as the C library are mapped into the process's virtual address space.
- **Stack** — contains local variables and function-call information.
- **VDSO/VVAR** — special kernel-provided mappings used for certain system operations and timing-related functionality.

The permissions shown in `/proc/<PID>/maps` describe what operations are allowed:

r = read

w = write

x = execute

p = private mapping

s = shared mapping

Result

The memory organization of a running Linux process was successfully studied using `/proc/<PID>/maps`, `/proc/<PID>/status`, `/proc/<PID>/smaps`, and `pmap`. The experiment demonstrated how **virtual memory mappings** divide a process's address space into code, data, heap, stack, libraries, and other regions.